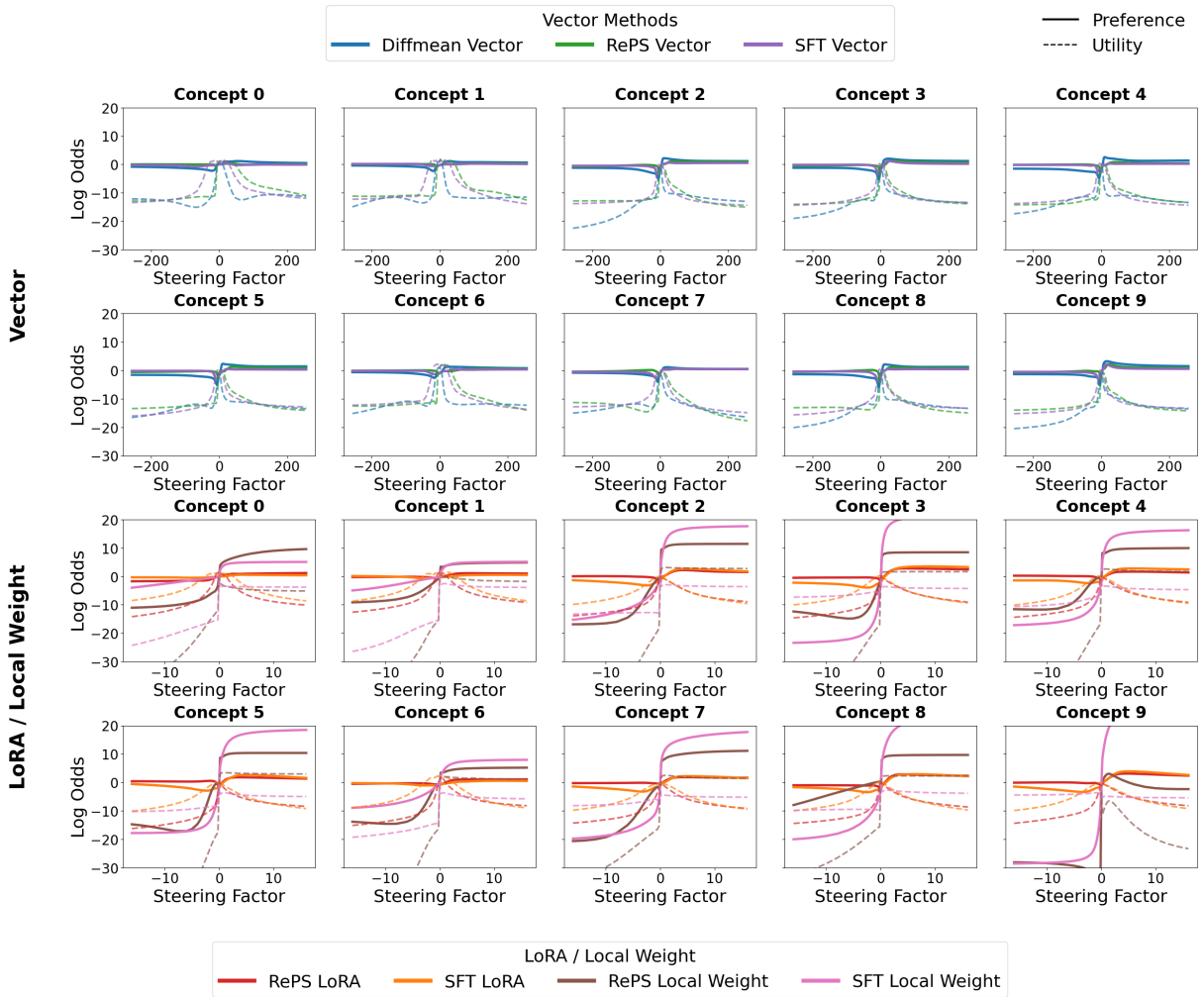


Figure 4: **Unified preference and utility dynamics under steering.** Solid lines represent preference log-odds, and dashed lines represent utility log-odds. The top panel shows steering with vector-form parameter modifications, and the bottom panel shows parametric interventions including LoRA and local weight updates. Results are shown for the `Qwen-2.5-7B-IT` model on the *AxBench* dataset, evaluated over its top 10 concept subsets. The horizontal axis corresponds to the steering factor.

C.3 Preference log-odds and Utility log-odds

Here we show how the preference and utility capability can be represented as (14) and (16). We take preference log-odds as example. First, the conditional probability before steering is given by:

$$\begin{aligned} P(p_p \mid h) &= \sigma\Big(-\boldsymbol{\omega}_p^{\mathsf{T}} h D_p(0) - b_p \Big), \\ &= \sigma(\eta) \end{aligned} \tag{29}$$

where $\eta \triangleq -\boldsymbol{\omega}_p^{\mathsf{T}} h D_p(0) - b_p$.

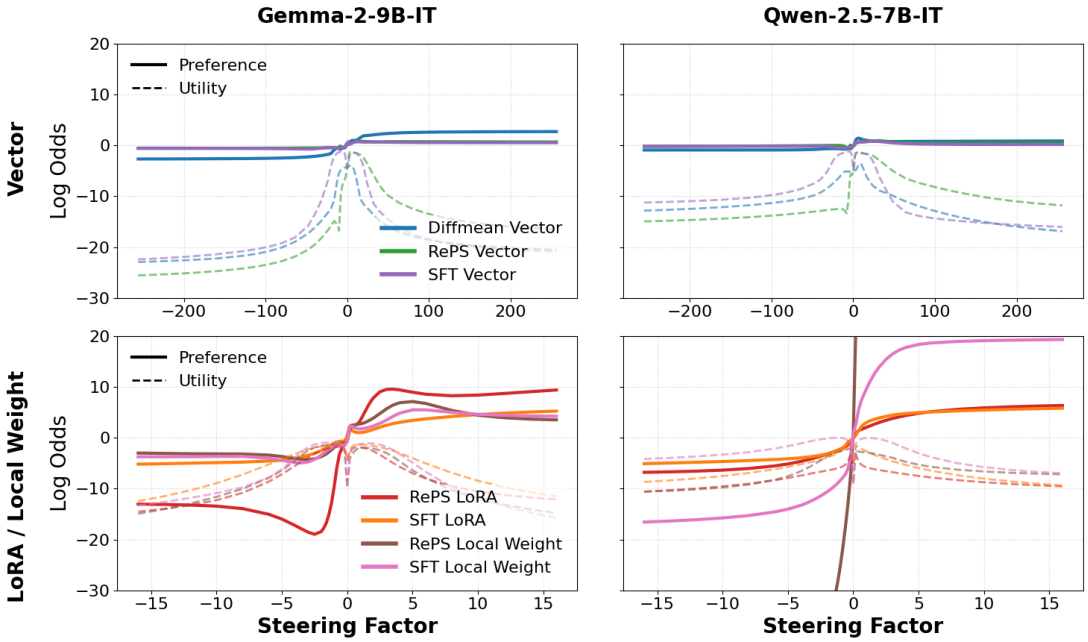
When an intervention at layer l updates the hiden state as $\tilde{h}(m) = h + m\Delta h$, we can get the steered preference probablity as (13) :

$$P(p_p \mid \tilde{h}(m)) = \sigma\left(-(\boldsymbol{\omega}_p^{\top} h + \alpha_p m) D_p(m) - b_p\right)$$

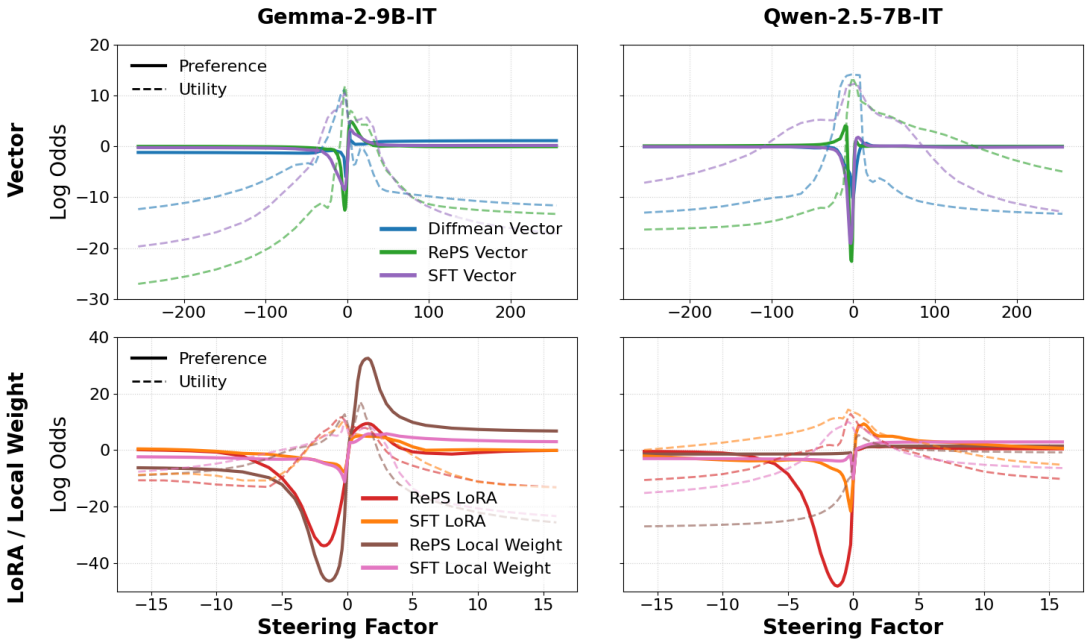
Next, we can represent the initial preference log-odds as $-\eta$:

$$\begin{aligned} \log \frac{P(p_p \mid h)}{P(p_n \mid h)} &= \log \frac{P(p_p \mid h)}{1 - P(p_n \mid h)} \\ &= \log \frac{\sigma(\eta)}{1 - \sigma(\eta)} \\ &= \log \frac{1/(1+e^{\eta})}{1 - 1/(1+e^{\eta})} \\ &= \log \frac{1/(1+e^{\eta})}{e^{\eta}/(1+e^{\eta})} \\ &= \log e^{-\eta} \\ &= -\eta \\ &= \boldsymbol{\omega}_p^{\mathsf{T}} h D_p(0) + b_p \end{aligned} \tag{30}$$

Finally, when we steering h by $\tilde{h}(m)$, we can get



(a) Powerseeking Results



(b) Psychopathy Results

Figure 5: **Unified preference and utility dynamics under steering.** Solid lines represent preference log-odds, and dashed lines represent utility log-odds. Figure (a) shows the unified preference and utility dynamics of the power-seeking dataset under two different models, while Figure (b) shows the results for the psychopathy dataset. The horizontal axis corresponds to the steering factor.

preference log-odds by:

$$\log \frac{P(p_p \mid \tilde{h}(m))}{P(p_n \mid \tilde{h}(m))} = -\eta_{steered} = (\boldsymbol{\omega}_p^\top h + \alpha_p m) D_p(m) + b_p \tag{31}$$

For utility capability, we have:

$$P(u \mid \tilde{h}(m)) = \sigma\Big(-\boldsymbol{\omega}_u^\top \tilde{h}(m) D_u(m) - b_u \Big). \tag{32}$$

For preference steering directions, we typically have $\boldsymbol{\omega}_u^\top \Delta h \approx 0$. So we can quantify utility capa-

bility by:

$$\log \frac{P(u \mid \tilde{h}(m))}{1 - P(u \mid \tilde{h}(m))} = \boldsymbol{\omega}_u^\top h D_u(m) + b_u$$

D Fitting Experiment Details

D.1 Fitting Results on Test Set

To further validate our theoretical model, we performed parameterized fitting on the test set using the SLSQP algorithm, strictly enforcing continuity between positive and negative segments at the origin. As shown in Table 5, the direct fitting yielded high goodness-of-fit values ($R^2 > 0.95$) for most methods. This confirms that the steering effect follows a deterministic trajectory predicted by our theory rather than random perturbations, thereby validating the proposed interaction mechanism.

D.2 Analysis of Generalization Ability

Following the validation of our theoretical mechanism, we conducted train-to-test transfer experiments to evaluate the extent to which different methods decouple "concepts" from specific inputs. Theoretical curve parameters were derived solely from training data and applied directly to the test set for prediction (Table 6).

Robust Generalization Overall, the fitted curves generalize well to held-out data, with vector-based interventions achieving consistently strong R^2 across most settings. Input-dependent approaches such as LoRA- and local-weight-based methods also generalize well in many cases, but exhibit larger variance across datasets and occasional failures, suggesting that input-dependent updates can be more sensitive to the evaluation distribution.